CO3- ASSESSMENT TOOL 2

SIMULATION TASK

1. FIR Quantization using Truncation

Simulate a 16-tap FIR low-pass filter with cutoff frequency 0.4π rad/sample using 8-bit fixed-

point coefficient truncation and analyze the resulting quantization error. (25) (BL3) (WK4)

Given Data:

- **FIR Filter Length = 16 taps**
- **Cutoff Frequency = 0.4π rad/sample**
- **Coefficient Word Length = 8 bits**
- **Input Signal = Sum of 500 Hz and 2 kHz sinusoids**
- **Sampling Frequency = 8 kHz**

Tasks:

1. **Design the ideal FIR low-pass filter.**
2. **Quantize the coefficients using truncation.**
3. **Simulate the filter response using MATLAB/Python.**
4. **Plot the magnitude responses before and after quantization.**
5. **Determine the output error signal.**
6. **Calculate Mean Square Error (MSE).**

Program

clc;

```
clear;
close all;


%% Given Data
N = 16;                          % FIR Filter Length
fc = 0.4;                        % Normalized cutoff
frequency (0.4*pi)
Fs = 8000;                       % Sampling Frequency (Hz)
bits = 8;                        % Coefficient Word Length


%% 1. Design the Ideal FIR Low-Pass Filter
h = fir1(N-1, fc, hamming(N));


%% Frequency Response Before Quantization
[H,f] = freqz(h,1,1024,Fs);


%% 2. Quantize Coefficients Using Truncation
scale = 2^(bits-1);


% Truncation
hq = fix(h * scale) / scale;


%% Frequency Response After Quantization
[Hq,f] = freqz(hq,1,1024,Fs);


%% 3. Input Signal
t = 0:1/Fs:0.02;                  % 20 ms signal


x = sin(2*pi*500*t) + sin(2*pi*2000*t);


%% Filter the Signal
y_original = filter(h,1,x);
y_quantized = filter(hq,1,x);
```

```matlab
%% 5. Output Error Signal
error_signal = y_original - y_quantized;


%% 6. Mean Square Error (MSE)
MSE = mean(error_signal.^2);


fprintf('Mean Square Error (MSE) = %f\n',MSE);


%% 4. Magnitude Response Plot
figure;


subplot(2,1,1)
plot(f,20*log10(abs(H)),'b','LineWidth',2);
hold on;
plot(f,20*log10(abs(Hq)),'r--','LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response');
legend('Original','Quantized');


%% Error Signal Plot
subplot(2,1,2)
plot(t,error_signal,'k','LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Output Error Signal');


%% Coefficient Comparison
figure;
stem(h,'b','filled');
hold on;
stem(hq,'r');
grid on;
xlabel('Coefficient Index');
ylabel('Coefficient Value');
title('Original vs Quantized FIR Coefficients');
legend('Original','Quantized');
```

```matlab
%% Output Signal Comparison
figure;

subplot(2,1,1)
plot(t,y_original,'b','LineWidth',1.5);
hold on;
plot(t,y_quantized,'r--','LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Filtered Output');
legend('Original','Quantized');

subplot(2,1,2)
plot(t,error_signal,'m','LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Error');
title(['Error Signal (MSE = ',num2str(MSE),')']);
clc;
clear;
close all;

%% Given Data
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Fs = 8000;                      % Sampling Frequency (Hz)
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hold on;
stem(hq,'r');
grid on;
xlabel('Coefficient Index');
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title('Original vs Quantized FIR Coefficients');
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hold on;
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grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Filtered Output');
legend('Original','Quantized');


subplot(2,1,2)
plot(t,error_signal,'m','LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Error');
title(['Error Signal (MSE = ',num2str(MSE),')']);
```
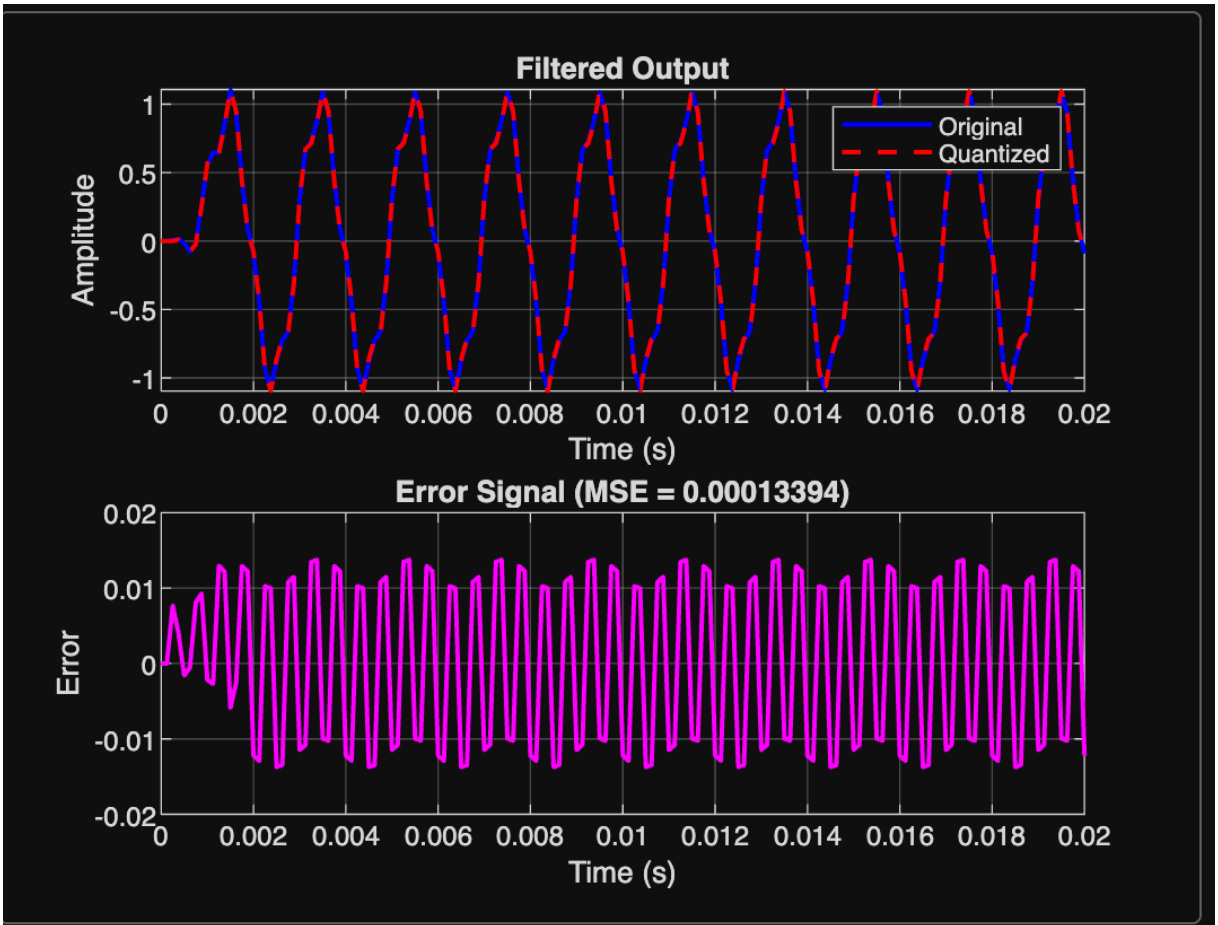
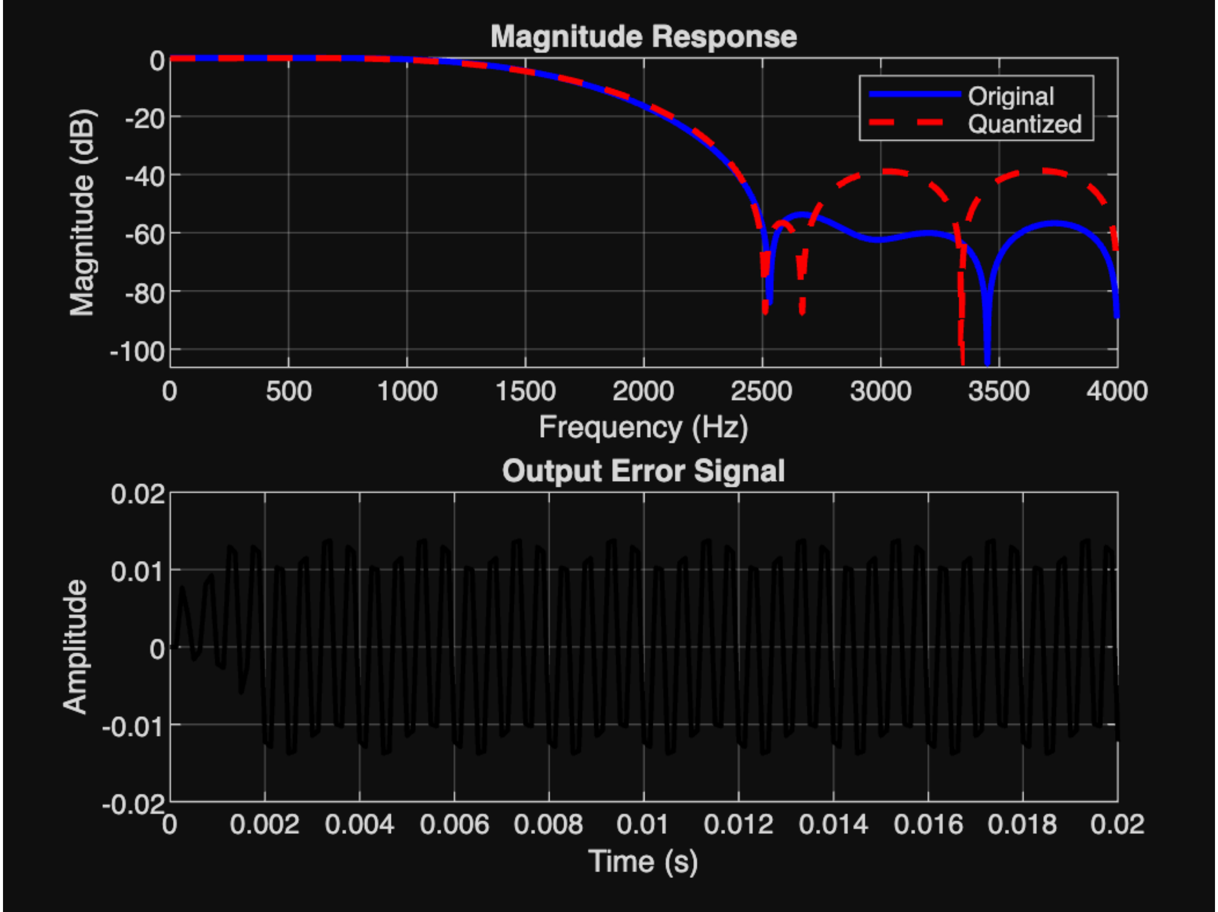
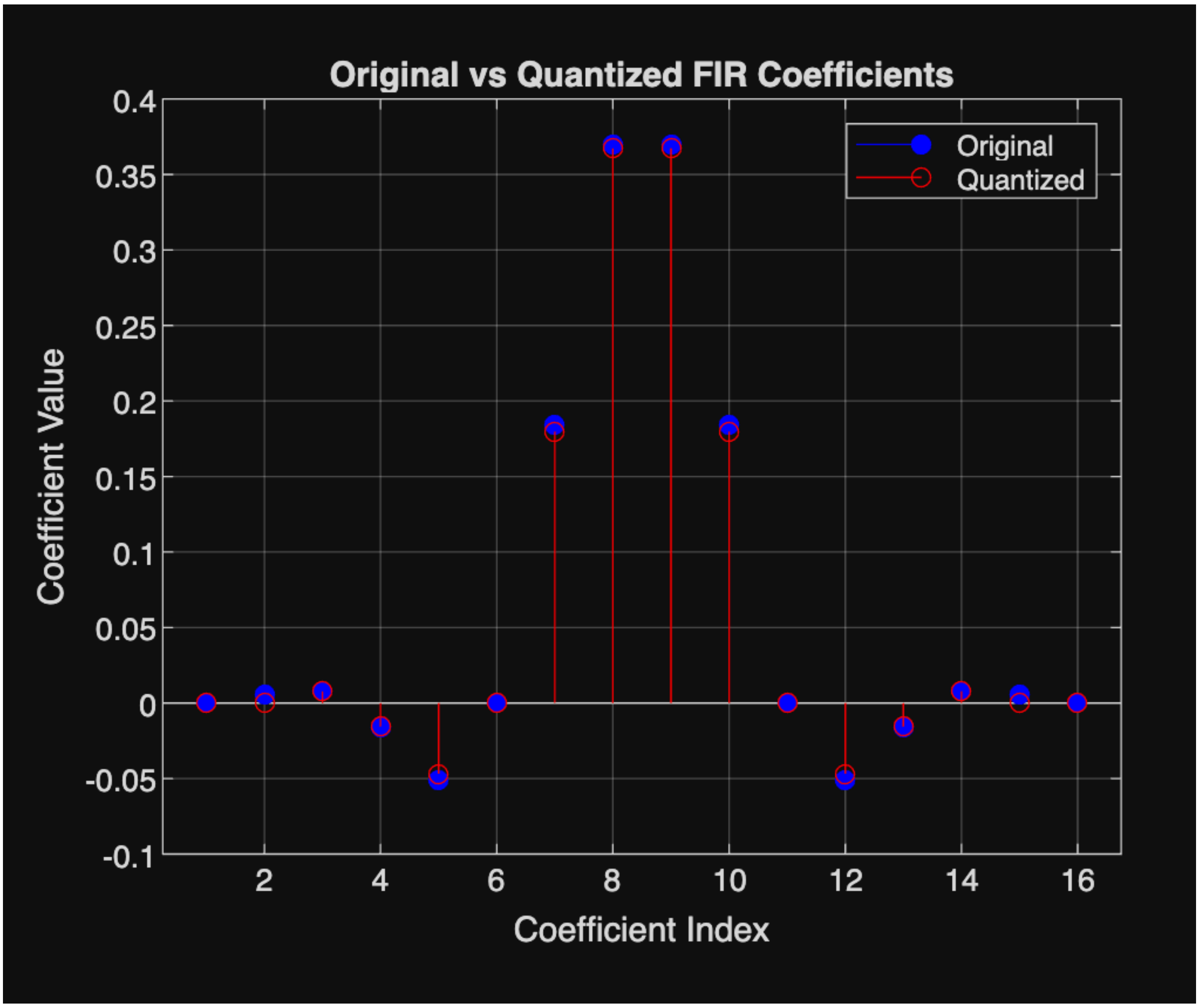




```
Warning: An error occurred while drawing the scene: Could not find node
in node tree during replaceCamera
```

```
Mean Square Error (MSE) = 0.000134
```




Original vs Quantized FIR Coefficients
Original
Quantized
Coefficient Value
Coefficient Index
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-0.05
-0.1
2
4
6
8
10
12
14
16